Protopasta
Filament by **Protoplant**

CFP1xxxx-ESD
Technical Data Sheet
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Protopasta **Static Dissipative Carbon Fiber PLA** is a Natureworks INGEO based PLA and carbon black/carbon fiber composite designed with the electronics fabrication and repair industry in mind. With easy printing and reliable resistivity, this material is perfectly suited to fixtures and other prints requiring extra stiffness, dimensional stability, and layer-hiding texture.

For quality control during production, we measure the resistance of 10 cm of 1.75 mm filament. We use a custom tool to apply a constant force where the filament contacts the 3 mm polished pin terminals. Below are our QC ranges for this test method for our conductive and dissipative materials.

Material Under Test	**Result Range**	**Meter Used**
Protopasta Conductive PLA	2.0 kΩ - 3.5 kΩ	Standard DMM
Protopasta Dissipative Materials	100 kΩ - 100,000 kΩ or (10^5 Ω - 10^8 Ω)	DESCO 19786

Formulated and compounded in-house, the electrical performance is within the dissipative range for ESD-safe materials as defined by ANSI/ESD S541 and IEC 613400 with surface resistance (resistivity) of 1 x 10^4 - 1 x 10^{11} ohms (1 x 10^5 - 1 x 10^{12} ohms/sq). With current processing and ingredients, our dissipative materials are typically 1 x 10^4 - 1 x 10^7 ohms (1 x 10^5 - 1 x 10^8 ohms/sq) surface resistance (resistivity).

To verify printed part surface resistance (resistivity) and to determine variation from orientation + print temperature, we printed 70 mm x 70 mm x 1.25 mm coupons. A Bambu P1S with AMS and .4 mm nozzle was used to print these parts at 12 mm^3/s with 100% infill. The resistance was measured using certified DESCO 19786 meter and DESCO 50005 Concentric Ring Probe.

Surface Resistance (Ω) of Printed Part (For Reference Only, Batch and Print Parameter Dependent)

Print Orientation	**Print Temperature (˚C)**	**Surface Resistance (Ω)**
Flat (X-Y)	210	1 x 10^5
Flat (X-Y)	230	1 x 10^5
Vertical (Z)	210	1 x 10^6
Vertical (Z)	230	5 x 10^5

Surface Resistivity (Ω/sq) of Printed Part (For Reference Only, Batch and Print Parameter Dependent)

Print Orientation	**Print Temperature (˚C)**	**Surface Resistivity (Ω/sq)**
Flat (X-Y)	210	1 x 10^6
Flat (X-Y)	230	1 x 10^6
Vertical (Z)	210	1 x 10^7
Vertical (Z)	230	5 x 10^6

Please note the above is an example only and results will vary depending on where in our QC range the batch tests, what your print and geometry parameters are, and your environmental conditions.

Recommended Printing Parameters

Print Temperature	210 - 230 °C
Build Plate temperature	60 °C
Print Speed (Volume Flow)	Up to 12 mm^3/s
Build Plate Preparation	PEI or similar w/ Magigoo or glue stick for release
Nozzle	.4 mm or larger
Abrasive	Abrasive. Hard nozzles recommended
Drying	Generally not required. 55 °C for 6 hrs
AMS Compatibility	Compatible, but wears Teflon and inline plastic parts

Processing Notes

Carbon black based conductive products are extremely dark and generally require more purge to change material than other colors.

Electrically dissipative polymer composites are sensitive to print orientation, temperature, infill, part geometry, and print speed. Users must test suitability for their specific application.

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